

Ghost Tool

User Manual — Version 3.3.0 (b4_ghost_tool_v7)

Ghost keyframe visualization and manipulation for **Bforartists**

See, grab, and reshape your in-betweens — directly in the 3D viewport.

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1. What is Ghost Tool?

Ghost Tool gives animators the ability to visualize and directly manipulate in-between frames in 3D space. It generates **ghost markers** at mathematically interpolated positions between your existing keyframes, displays them as draggable handles in the viewport, and recalculates f-curves in real time as you move them. Instead of bouncing between the Graph Editor and the viewport, you sculpt your motion where you can see it.

Core ideas

Ghosts — markers at interpolated positions between keys. Keyframe ghosts sit on actual keys; in-between ghosts show what interpolation does between them. Dragging an in-between reshapes the curve.

Per-scene sessions — settings and ghost data live on each scene (*scene.ghost_tool*). The first time you use a scene, press **Initialize Ghost Tool** in the panel.

Non-destructive preview, explicit commit — overlays (trails, onion skin, physics previews, visual diff) never touch your keys. Only explicit actions (drag confirm, Stamp to Keys, easing apply, snapshot restore) write f-curves, and all of them are undoable.

Requirements & installation

Bforartists 4.x (the addon is **Bforartists-exclusive**; on standard Blender it shows only an information panel). Install via *Preferences* → *Add-ons* → *Install from Disk* and choose `b4_ghost_tool_v7.zip`, then enable **Ghost Tool**. The panel appears in the 3D viewport sidebar (N) under the **Ghost Tool** tab.

2. Five-minute quick start

	Step
1	Open the Ghost Tool tab in the 3D viewport sidebar and click Initialize Ghost Tool (first use per scene).
2	Select an animated object (or armature in Pose mode with a bone selected) and click Generate Ghosts .
3	Hover a ghost and drag it with Shift+G (or the Drag Ghost button). Constrain with X/Y/Z, or Shift+X/Y/Z for planes. Left-release confirms, right-click/Esc cancels.
4	Watch the f-curve update live. Undo works normally after confirming.
5	Try Take Snapshot before experiments — restore any saved state later from Saved Snapshots.

Hotkeys: **Shift+G** drag ghost · **Shift+B** box-select ghosts (Pose and Object mode).

3. Panel reference

All panels live under the root **Ghost Tool** panel in the sidebar. Advanced panels start collapsed.

Panel	What it does
Ghost Tool (root)	Master enable toggle, Initialize button, generate/clear ghosts, status readouts. If initialization or drawing ever fails, this panel shows the error instead of breaking the sidebar.
Onion Skin	Classic ghosting of the active object across neighbouring frames for spacing reference.

Panel	What it does
Motion Trails	Curve-style trajectory lines through ghost positions, so arcs read at a glance.
Marker Placement	Controls how many in-betweens are generated and where: subdivision level, curve mode, frame ranges.
Marker Display	Visual styling: ghost radius, opacity, per-state colors (keyframe / in-between / pinned / selected), level colors. Starts collapsed.
Marker Tools	The workbench: drag and selection tools, easing presets, pin/unpin, quick snapshot row, Physics Suggest, Physics Feel archetype generator, and Visual Diff (new in 3.3).
Export / Import	Save the entire ghost configuration to JSON or load one (positions, pins, settings). Great for sharing setups between files or artists.
Settings	Behavior knobs and performance options. Starts collapsed.
Saved Snapshots	Every snapshot you take, with restore / toggle-visibility / delete controls.
Floating Toolbar	Pops the main strip out as a movable dialog (button in panel header area) for dual-monitor or fullscreen workflows.

4. Working with ghosts

Generating

Generate Ghosts samples the active object's f-curves (or the selected pose bones) and places markers between keys according to Marker Placement settings. **Clear All Ghosts** removes them. Regenerate after large timing edits — the pipeline also refreshes automatically after drags via a lightweight deferred update.

Dragging & selection

Drag Ghost (Shift+G) casts a ray from your cursor and slides the ghost on its constraint plane. Axis locks: press X, Y or Z during the drag; Shift+axis locks the perpendicular plane; press again to release. Confirm with left-release; cancel with right-click or Esc — cancel restores the f-curves exactly from an internal snapshot. **Select Ghost**, **Box Select** (Shift+B) and hover highlighting handle multi-ghost workflows.

Pinning

Pin Marker toggles a pin on the ghost nearest your cursor; pinned ghosts are protected reference points (shown with the pinned color and counted in the panel). **Unpin All** releases them.

Easing presets

Apply curve shapes to the selected ghost range: **Ease In**, **Ease Out**, **Ease In-Out**, **Linear**, **Overshoot**, **Anticipate**, **Bounce**, or **Custom**. This rewrites interpolation across the selected span in one click.

Snapshots

Take Snapshot stores the current ghost configuration; **Restore Snapshot** writes f-curves back to match it; **Toggle Snapshot Visibility** overlays it for comparison; **Delete Snapshot** removes it. Use snapshots as cheap versioning while exploring alternatives.

Mesh ghosts

Generate Mesh Ghosts creates transparent duplicates of the mesh at frames around the playhead (true silhouette onion-skinning); **Toggle Mesh Ghost Mode** cycles Solid → Wireframe → Off; **Clear Mesh Ghosts** removes them.

5. Physics Feel — archetype generator (new in 3.3)

Physics Feel stamps physically-plausible motion onto a channel without simulation. Pick an archetype, preview the displacement overlay in the viewport, then write real keyframes with one button. All curves are pure math (no AI, no physics engine) and respect two hard guarantees: they start at zero displacement, and return-to-rest archetypes end at **exactly** zero so the last stamped key can never pop.

Archetype	Shape	Behavior
PENDULUM	Damped sine swing	Decaying oscillation, ~1.25 cycles. Ends exactly at rest (endpoint corrected in v3.3).
BOUNCE	Parabolic bounces	4 bounce arcs with 0.55 height decay. Floors at zero between arcs.
SETTLE	Exponential ease-in	Asymptotic approach to the target; reaches ~99.3% at the end frame.
WOBBLE	Fast damped jitter	4-cycle oscillation under a tight decay envelope; reads as secondary motion.
SPRINGBACK	Overshoot & return	Peaks ~1.4× past the target near 60% of the range, then returns to rest.

Workflow

1) In **Marker Tools** → **Physics Feel**, choose the archetype and switch on **Preview** — a cyan displacement overlay appears. 2) Set **Amplitude** (world units) and **Axis** (X/Y/Z location channel). 3) Set the **Start/End** frame range. 4) Choose what happens to existing keys in range (**REPLACE** clears them; **OFFSET** is reserved for a future version and shown greyed out). 5) Press **Stamp to Keys** — one key per frame is written to the chosen channel.

Stamping requires an existing location keyframe on the target axis (the f-curve must exist), and refuses to run while NLA strips are active on the object — mute or exit NLA first. The stamped result is plain keyframes: fully editable, fully undoable.

6. Visual Diff Mode (new in 3.3)

Visual Diff answers "what actually changed?" while polishing poses. Pin a reference frame; every bone is then tinted on a **cool** → **warm** gradient by how far it has moved (world-space) from that reference pose. The overlay is purely cosmetic — it never writes keys.

Using it

In **Marker Tools** → **Visual Diff**: jump to the pose you want as reference and press **Pin Reference** (stores the current frame's bone positions and enables the overlay). Scrub or pose; bones light up warm as they diverge. **Max Distance** sets how far equals "fully warm" (default 0.5 units). The two color swatches customize the

cool/warm ends. **Unpin** clears the reference and hides the overlay.

Staleness Guard

The pin stores a hash of the anchor. If the underlying data changes so the reference no longer matches (keys edited on the anchor frame, rig changed), the overlay desaturates and the HUD mode label warns **STALE** — re-pin to refresh. This prevents silently comparing against an outdated pose.

Requires an armature (it reads pose-bone world positions from the evaluated depsgraph).

7. Physics Suggest — arc correction

Physics Suggest (Preview) analyses the selected ghosts' trajectory and proposes a parabolic correction — useful for ballistic arcs (jumps, throws) that hand-keying flattened. Preview shows the corrected arc in the viewport; applying moves the ghosts (and therefore the keys) onto it. Like everything else: preview is free, apply is undoable.

8. Operator reference (24 operators)

Every button in the UI maps to one of these operators. They are also callable from scripts as `bpy.ops.<idname>`, and searchable via F3.

Identifier	Label	Description
<code>ghost_tool.apply_easing</code>	Apply Easing Preset	Apply an easing preset to the selected ghost range
<code>ghost_tool.archetype_bake</code>	Stamp to Keys	Stamp archetype-shaped keyframes onto the active object's fcurve
<code>ghost_tool.box_select</code>	Box Select Ghosts	Draw a selection rectangle to select multiple ghost markers
<code>ghost_tool.clear_ghosts</code>	Clear All Ghosts	Remove all ghosts from the current scene
<code>ghost_tool.clear_mesh_ghosts</code>	Clear Mesh Ghosts	Remove all transparent mesh ghost duplicates from the scene
<code>ghost_tool.delete_snapshot</code>	Delete Snapshot	Delete a snapshot permanently
<code>ghost_tool.drag_ghost</code>	Drag Ghost	Modal operator for dragging ghost markers in the 3D viewport
<code>ghost_tool.export_ghosts</code>	Export Ghosts	Export ghost configuration to a JSON file
<code>ghost_tool.floating_toolbar</code>	Ghost Tool — Floating Toolbar	Pop out the Ghost Tool strip as a floating toolbar dialog
<code>ghost_tool.generate_ghosts</code>	Generate Ghosts	Generate ghost markers for the active object and selected bones
<code>ghost_tool.generate_mesh_ghosts</code>	Generate Mesh Ghosts	Create transparent mesh duplicates at frames around the playhead to visualize the character pose at different times
<code>ghost_tool.hover_detect</code>	Ghost Hover Detect	Modal operator that runs a lightweight timer to track which ghost
<code>ghost_tool.import_ghosts</code>	Import Ghosts	Import ghost configuration from a JSON file

Identifier	Label	Description
ghost_tool.initialize	Initialize Ghost Tool	Attach Ghost Tool settings to this scene
ghost_tool.physics_suggest	Physics Suggest (Preview)	Preview and apply parabolic arc correction to ghosts (physics suggestion)
ghost_tool.pin_diff_reference	Pin Diff Reference	Pin the current frame as the Visual Diff reference. Bones will be colored warm/cool by their distance from this pose
ghost_tool.pin_ghost	Pin/Unpin Ghost	Toggle pin state on the ghost nearest to the cursor
ghost_tool.restore_snapshot	Restore Snapshot	Restore f-curves to match a selected snapshot
ghost_tool.select_ghost	Select Ghost	Select or toggle-select the ghost under the cursor
ghost_tool.take_snapshot	Take Snapshot	Take a snapshot of the current ghost configuration
ghost_tool.toggle_mesh_mode	Toggle Mesh Ghost Mode	Cycle through mesh ghost display modes: Solid, Wireframe, Off
ghost_tool.toggle_snapshot	Toggle Snapshot Visibility	Toggle the visibility of a snapshot overlay
ghost_tool.unpin_all	Unpin All Ghosts	Unpin all pinned ghosts in the current scene
ghost_tool.unpin_diff_reference	Unpin Diff Reference	Clear the pinned Diff Reference and hide the diff overlay

9. Scripting API

ghost_tool.api exposes a stable surface for pipeline tools: `get_ghost_positions()`, `set_ghost_position(uid, pos)` (recalculates f-curves), `refresh_ghosts()`, `get_ghost_count()`, `get_ghost_by_uid(uid)`, plus `on_ghost_moved(callback)` / `remove_callback()` for reacting to artist edits (callbacks are weakly referenced and cleared automatically when the addon unloads).

10. Troubleshooting

Symptom	Resolution
Panel says 'Ghost Tool not initialized'	Press Initialize Ghost Tool. Settings are per-scene; every new scene needs one click.
'Stamp to Keys' warns about NLA	The object has active NLA strips. Mute/stash them or exit NLA tweak mode, then stamp.
'Add at least one location keyframe on that axis first'	Physics Feel writes into an existing f-curve. Key the target axis once, then stamp.
Diff colors suddenly washed out + STALE label	Staleness Guard fired: the reference no longer matches stored data. Pin Reference again.
'No armature bones found – cannot pin Diff Reference'	Visual Diff reads pose bones. Select/Enter an armature-driven character.
Dragging does nothing	Make sure ghosts exist (Generate Ghosts), you're hovering one, and the viewport is active. Shift+G starts the modal drag.
Overlay feels heavy in dense scenes	Lower subdivision level, shrink the frame range, reduce trail/onion counts in their panels, or disable overlays you're not using.

Symptom	Resolution
Addon shows only an info panel	You're on standard Blender. Ghost Tool v7 is Bforartists-exclusive by design.
Buttons error after upgrading the zip	Restart Bforartists. The addon purges stale <code>__pycache__</code> on registration, but a restart guarantees a clean reload.

11. What's new in v3.3.0 (b4_ghost_tool_v7)

- **Visual Diff Mode** ported to the Bforartists build (pin reference, warm/cool overlay, Staleness Guard) — and given a proper panel section; upstream it existed but had no UI.
- **Physics Feel archetypes** ported: PENDULUM, BOUNCE, SETTLE, WOBBLE, SPRINGBACK with preview overlay and Stamp to Keys.
- **PENDULUM endpoint fix**: the final stamped key now lands exactly at rest (previously up to 3% of amplitude off, which could read as an end pop).
- Bforartists-exclusive gate restored (standard Blender gets an informational stub).
- Full diagnostic pass: registration order, handler/keymap hygiene, idname uniqueness, UI cross-references — clean.

Companion file for AI assistants: `GhostTool_LLM_Reference.md` in this folder contains the machine-readable deep reference (file map, invariants, full property tables).